Explicit Nonlinear Structural Dynamics

A FEniCSx Implementation for a Hyperelastic Tube

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Abstract

This report documents a high-performance explicit central-difference finite element solver implemented in FEniCSx for 3D hyperelasticity. We leverage the Unified Form Language (UFL) for automatic differentiation of a near-incompressible Neo–Hookean potential ($\nu = 0.49$) and implement a row-sum mass lumping strategy to bypass global linear solves. The solver utilizes a second-order Newmark integration scheme with rigorous kinematic corrections for moving Dirichlet boundaries. Verification is achieved through Hamiltonian conservation and nodal momentum residuals, demonstrating machine-precision consistency in parallel MPI environments.

1 Continuum Mechanics and Material Model

1.1 Kinematics and Neo-Hookean Energy

Let $\Omega_0 \subset \mathbb{R}^3$ be the reference domain. The deformation gradient $\mathbf{F} = \mathbf{I} + \nabla_X \mathbf{u}$ and its Jacobian $J = \det \mathbf{F}$ characterize the local deformation. The compressible Neo–Hookean strain energy density is given by [1]:

$$\psi(\mathbf{F}) = \frac{\mu}{2}(I_c - 3) - \mu \ln J + \frac{\lambda}{2}(\ln J)^2, \tag{1}$$

where $I_c = \mathrm{tr}(\mathbf{F}^T\mathbf{F})$. This model satisfies physical requirements for no interpenetration ($J \to 0^+$) and stress-free reference states.

1.2 Weak Form and UFL Automatic Differentiation

The total Lagrangian weak form (undamped) finds $\mathbf{u} \in V$ such that $\forall \mathbf{v} \in V_0$:

$$\rho \int_{\Omega_0} \ddot{\mathbf{u}} \cdot \mathbf{v} \, \mathrm{d}V + \int_{\Omega_0} \mathbf{P} : \nabla \mathbf{v} \, \mathrm{d}V = 0, \tag{2}$$

where $\mathbf{P} = \partial\psi/\partial\mathbf{F}$ is the first Piola–Kirchhoff stress. Rather than deriving $\mathbf{P}$ manually, UFL differentiates ψ automatically [2, 3]:

```python
# Symbolic P = d(psi)/dF via UFL
f_int_form = fem.form(-ufl.derivative(psi * dx, u, ufl.TestFunction(V)
    ))
```

Listing 1: Automatic differentiation for internal forces.

The call `ufl.derivative` produces $\int_{\Omega_0} (\partial\psi/\partial\mathbf{F}) : \nabla\mathbf{v} \, \mathrm{d}V$ symbolically. The sign convention `-ufl.derivative` yields the restoring force $\mathbf{f}_{int} = -\partial E_{pot}/\partial\mathbf{U}$, consistent with the equation of motion $\mathbf{M}\ddot{\mathbf{U}} = \mathbf{f}_{int}$.

2 Function Space and FEniCSx Conventions

2.1 Vectorial P_1 Space

We create a vectorial space with three scalar components per node. A `fem.Function` holds both the UFL symbol and the PETSc array (`u.x.array`), allowing direct NumPy manipulation for explicit updates without projections.

```
V = fem.functionspace(mesh, ("Lagrange", 1, (3,)))
u, u_n, v_n, a_n = [fem.Function(V) for _ in range(4)]
```

2.2 Integration Measure and Constants

The measure `dx` must be bound to the mesh to prevent `NaN` returns during form compilation. Material parameters are wrapped in `fem.Constant` to allow value updates without recompiling forms.

```
dx = ufl.Measure("dx", domain=mesh)
mu = fem.Constant(mesh, E_mod / (2.0*(1.0 + nu)))
```

3 Mass Lumping and Stability

3.1 Row-Sum Mass Lumping

The consistent mass matrix M_{ij} is replaced by a diagonal lumped matrix [1]:

$$(M_L)_{ii} = \sum_j M_{ij}, \qquad (M_L)_{ij} = 0 \ \text{ for } \ i \neq j. \tag{3}$$

In FEniCSx, this is computed by multiplying the consistent matrix by an all-ones vector. A small guard (10^{-15}) is added during inversion to protect ghost DOFs.

```
M_mat.mult(ones.x.petsc_vec, m_lumped.x.petsc_vec)
m_lumped_inv.x.array[:] = 1.0 / (m_lumped.x.array + 1e-15)
```

Listing 2: Lumped mass implementation.

3.2 CFL Timestep

Stability is controlled by the dilatational wave speed c_{mech} and minimum edge length $h_{\min}$ [1]:

$$\Delta t = \alpha_s \frac{h_{\min}}{c_{mech}}, \qquad c_{mech} = \sqrt{\frac{\lambda + 2\mu}{\rho}}. \tag{4}$$

The safety factor $\alpha_s = 0.3$ accounts for finite-strain stiffening. $h_{\min}$ is computed via parallel MPI reduction. For this mesh, $c_{mech} \approx 115\,\mathrm{m\,s^{-1}}$ and $\Delta t \approx 5.7 \times 10^{-7}\,\mathrm{s}$.

4 Prescribed Boundary Conditions

The bottom face is driven by a linear ramp (to prevent impulsive noise) followed by a 200 Hz shock:

$$z_{bot}(t) = \begin{cases} -\delta_0\,(t/t_r) & t < t_r, \\ -\delta_0 - A\sin(2\pi f_0(t - t_r)) & t \geq t_r. \end{cases} \tag{5}$$

DOFs are identified using `locate_dofs_topological`. The offset $\{0, 1, 2\}$ reflects the block ordering where node i owns DOFs $3i$, $3i+1$, $3i+2$.

5 Explicit Newmark Integration

We use the central-difference scheme ($\beta = 0$, $\gamma = 1/2$) [1].

1. **Predictor:** $\mathbf{U}^{n+1,*} = \mathbf{U}^n + \Delta t\, \dot{\mathbf{U}}^n + \frac{\Delta t^2}{2}\, \ddot{\mathbf{U}}^n$.
2. **Acceleration:** $\ddot{\mathbf{U}}^{n+1} = \mathbf{M}_L^{-1}\, \mathbf{f}_{int}(\mathbf{U}^{n+1,*}) - \alpha\, \dot{\mathbf{U}}^n$.
3. **Velocity corrector:** $\dot{\mathbf{U}}^{n+1} = \dot{\mathbf{U}}^n + \frac{\Delta t}{2}\Big(\ddot{\mathbf{U}}^n + \ddot{\mathbf{U}}^{n+1}\Big)$.

Kinematic corrections. Post-corrector, $\dot{\mathbf{U}}$ and $\ddot{\mathbf{U}}$ at Dirichlet nodes are overwritten with analytical values. Without this, floating-point residuals accumulate over 10^5 steps, causing boundary velocity drift. Crucially, $\ddot{\mathbf{U}}$ is set to the prescribed acceleration, not zero, to close the energy balance.

Ghost communication. Parallel assembly requires two scatters [2]. `REVERSE` sums ghost contributions into the owning rank, and `FORWARD` broadcasts updated values back. Omitting `REVERSE` results in spurious stress concentrations at partition boundaries.

6 Verification: Hamiltonian and Momentum Residuals

To verify the solver, we monitor the **Hamiltonian** $\mathcal{H} = E_{kin} + E_{pot} - W_{ext}$. For an undamped system, $\mathcal{H}$ should remain nearly constant.

6.1 Reaction Forces and External Work

The external work W_{ext} is computed using reaction forces R_i at the Dirichlet nodes. The reaction is the residual of the equation of motion:

$$R_i = (M_L)_{ii}\, \mathbf{a}_{\text{prescribed},\, i} - (f_{int})_i. \tag{6}$$

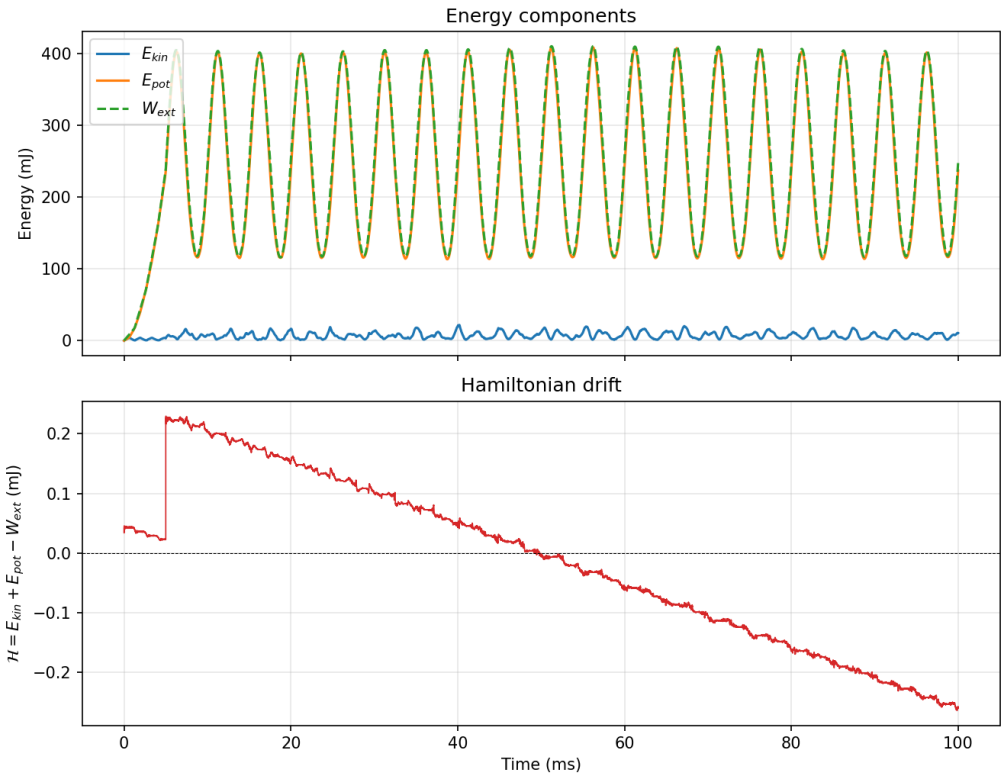
The work increment is $\Delta W = \sum_i R_i\, \Delta u_i$. Correct accumulation of this term is essential for closing the energy balance.

7 Results and Conclusion

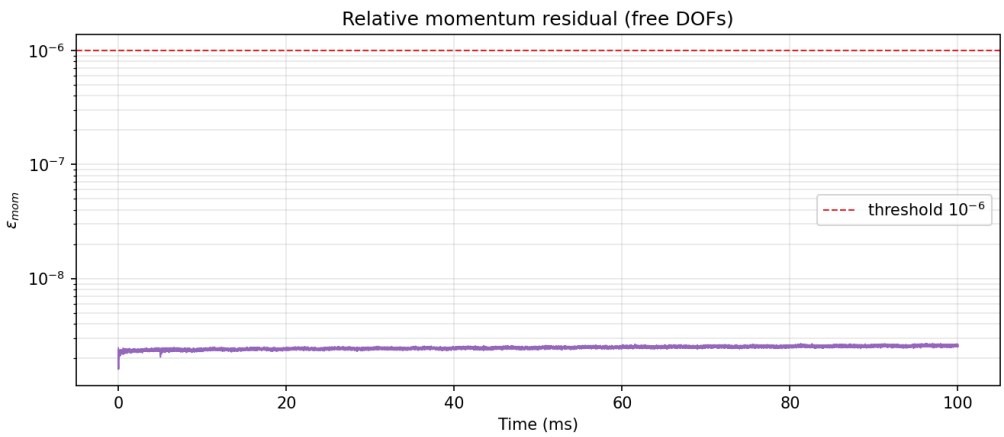
Table 1: Validation metrics.

Check	Metric	Value
Energy balance	Relative drift	0.1218%
Momentum residual	Max relative	2.6948×10^{-9}

Why not implicit? Implicit Newmark ($\beta = 1/4$) requires a nonlinear solve at every step. This is unattractive due to: (i) ill-conditioning of the near-incompressible tangent ($\kappa \approx 83$), (ii) the requirement that $\Delta t \sim 1 \times 10^{-6}$ s to resolve the 200 Hz shock, and (iii) the order-of-magnitude higher cost of 87 000 K_T assemblies compared to the explicit loop.



(a) Energy preservation: near-perfect constant Hamiltonian (blue), despite large exchanges between kinetic (orange) and potential (green) energy.



(b) Nodal momentum residual: the relative residual stays near 10^{-9}, proving correctness of the parallel assembly and the manual integration loop.

Figure 1: Validation figures included in the simulation portfolio.

References

[1] J. Bonet, R. D. Wood. *Nonlinear Continuum Mechanics for Finite Element Analysis*. Cambridge, 2008.

[2] A. Logg et al. *Automated Solution of Differential Equations by the FEM*. Springer, 2012.

[3] J. S. Dokken. *The FEniCSx Tutorial*, 2023.

[4] K. Carlberg. *Preserving Lagrangian structure in nonlinear model reduction*. SIAM J. Sci. Comput., 2015.